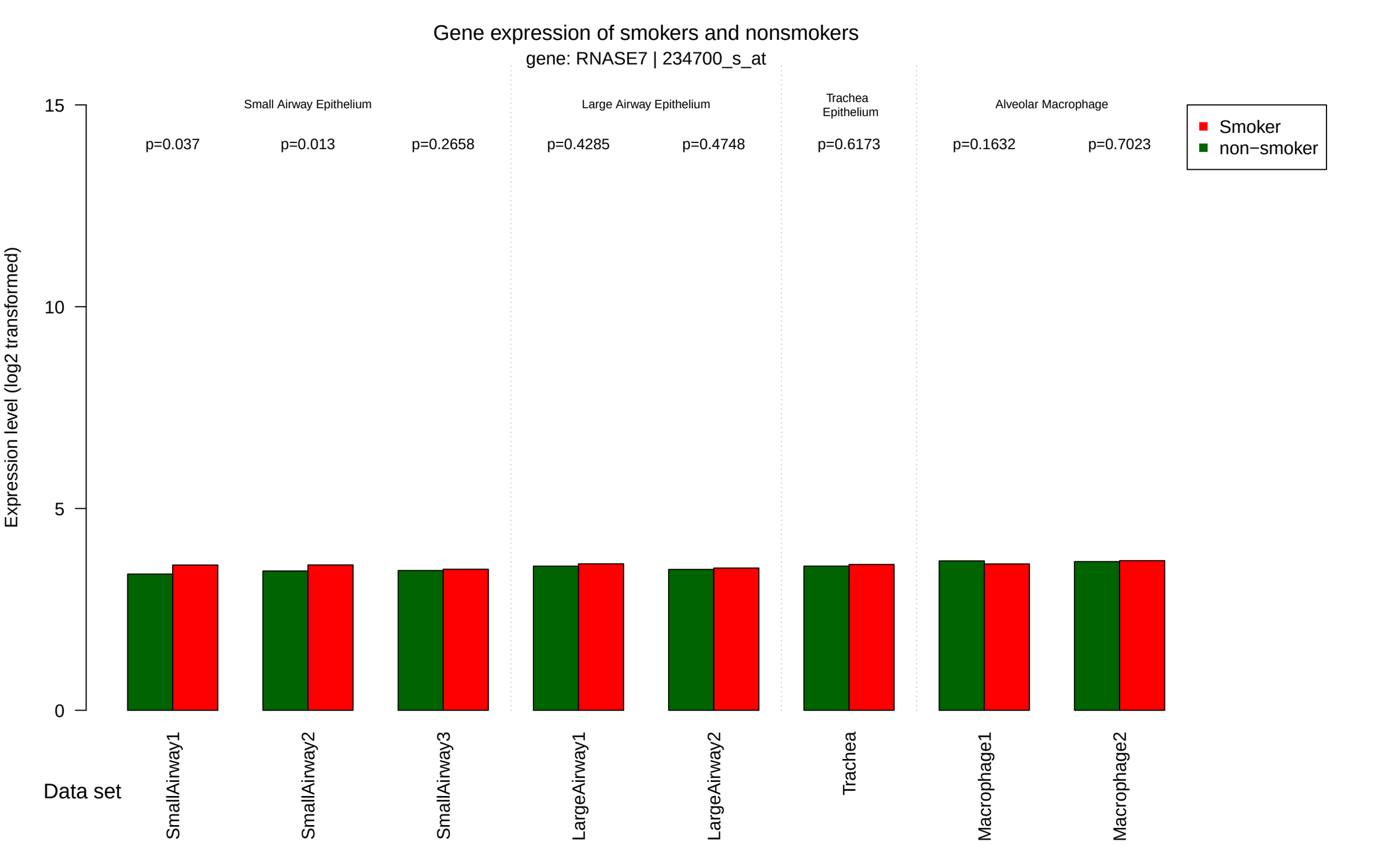
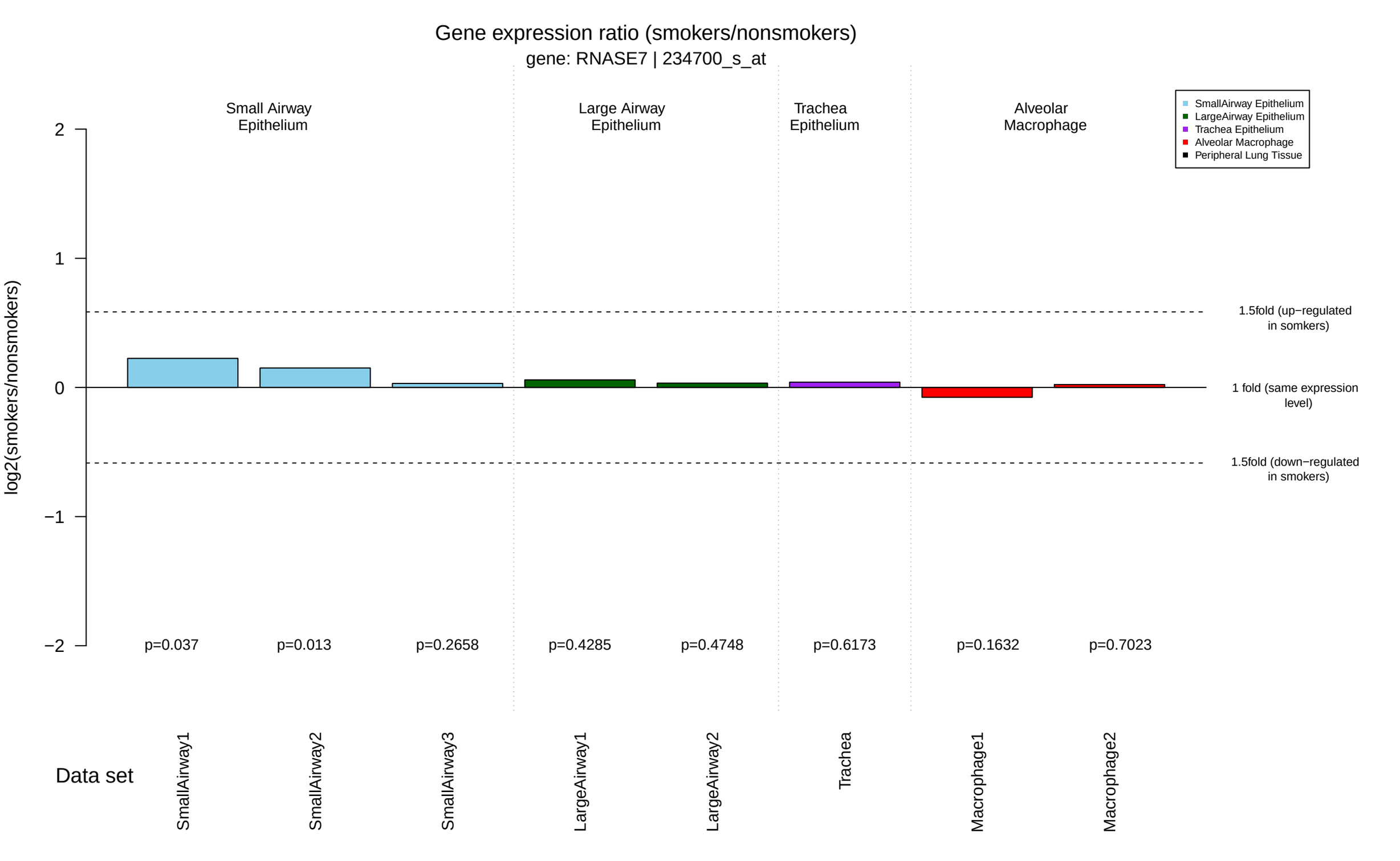
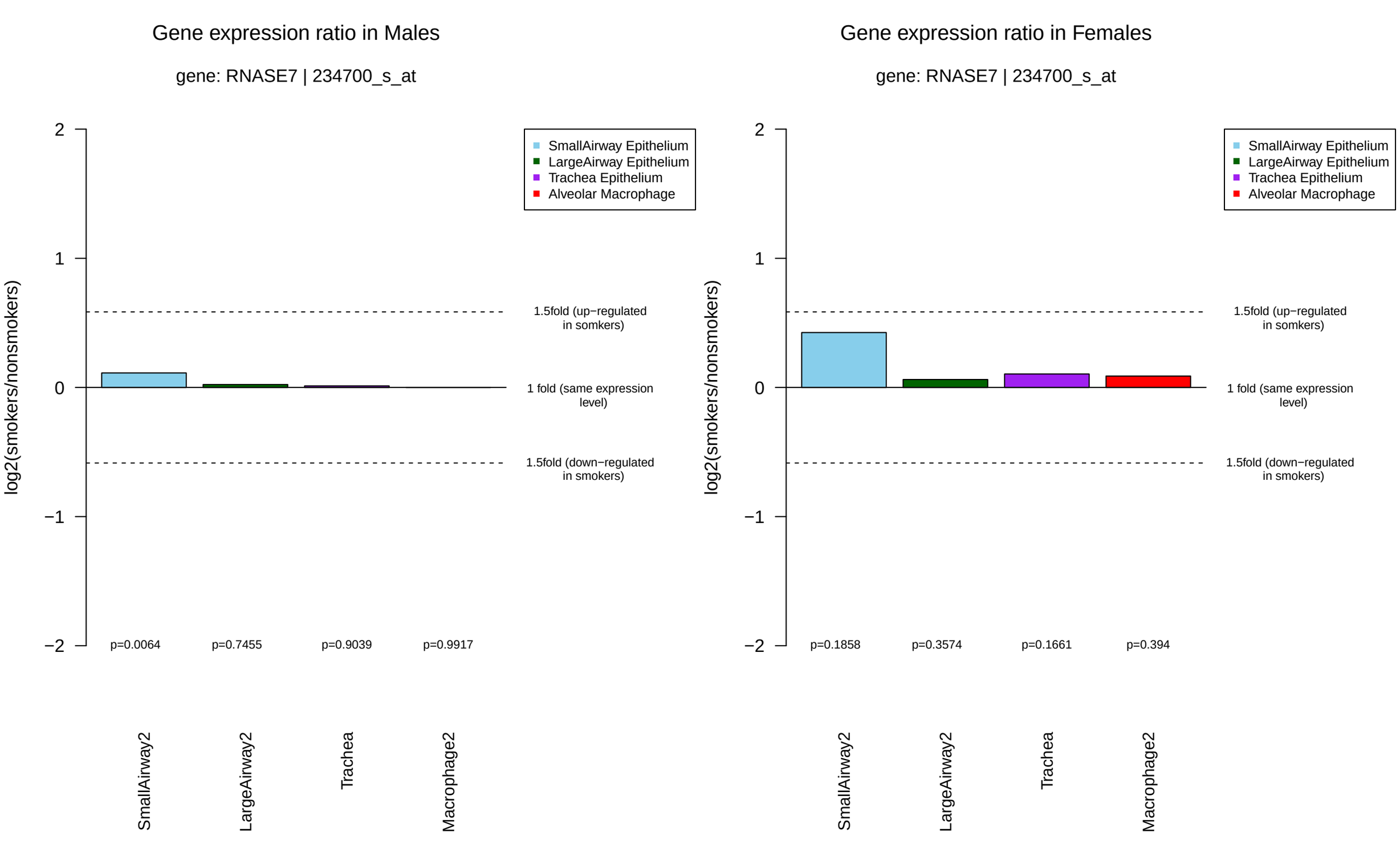
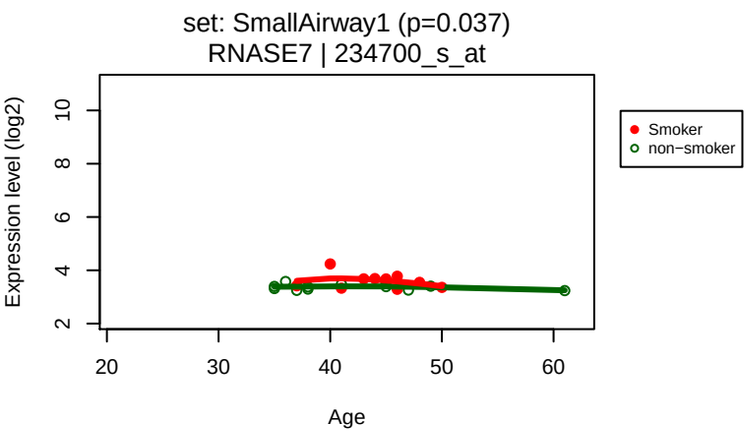
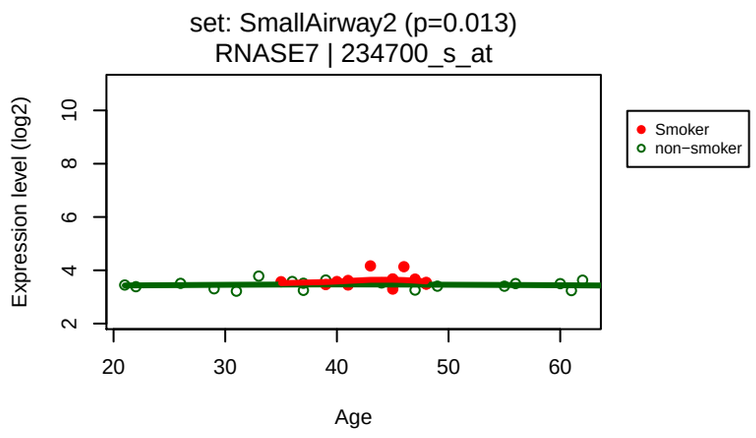
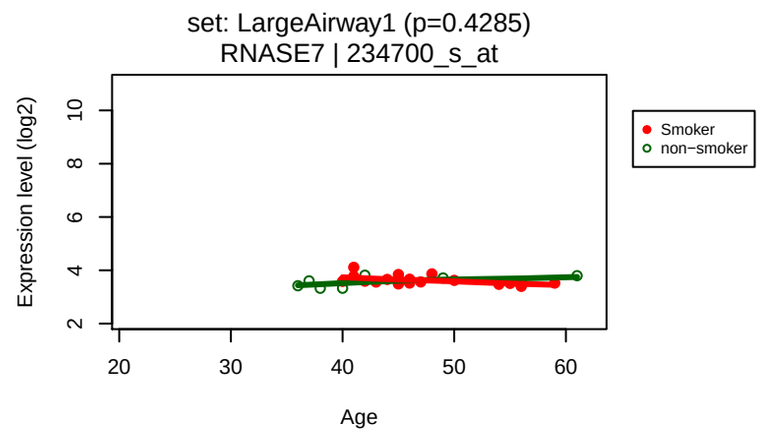

Gene expression of smokers and nonsmokers
gene: RNASE7 | 234700_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.037
p=0.013
p=0.2658
p=0.4285
p=0.4748
p=0.6173
p=0.1632
p=0.7023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

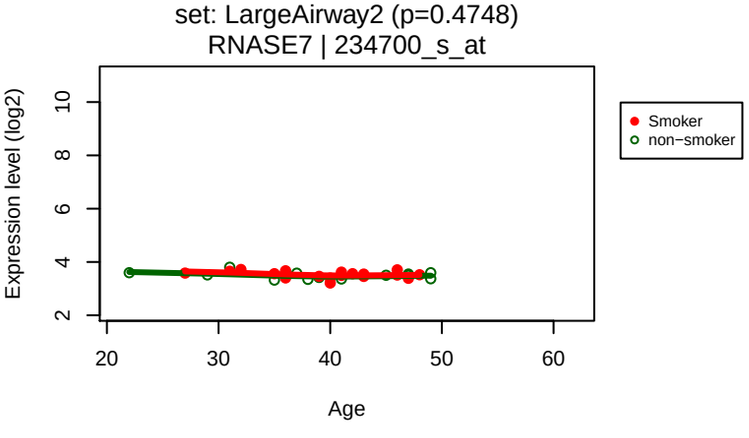

Gene expression ratio (smokers/nonsmokers)
gene: RNASE7 | 234700_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.037
p=0.013
p=0.2658
p=0.4285
p=0.4748
p=0.6173
p=0.1632
p=0.7023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

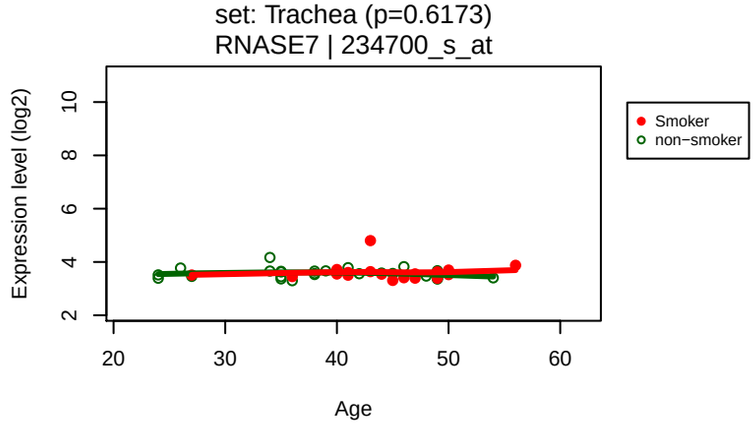

Gene expression ratio in Males
gene: RNASE7 | 234700_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0064
p=0.7455
p=0.9039
p=0.9917
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNASE7 | 234700_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1858
p=0.3574
p=0.1661
p=0.394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

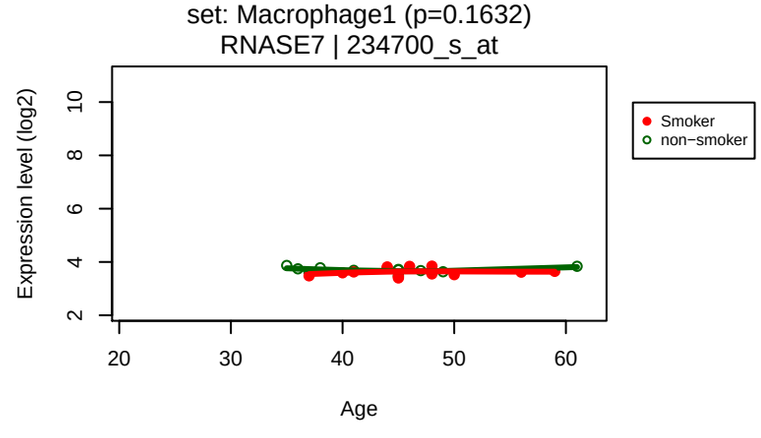

set: SmallAirway1 (p=0.037)
RNASE7 | 234700_s_at
Expression level (log2)
Age
Smoker
non−smoker

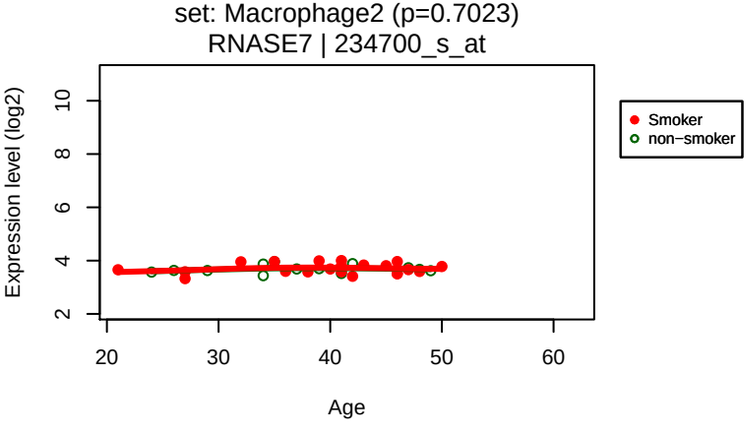

set: SmallAirway2 (p=0.013)
RNASE7 | 234700_s_at
Expression level (log2)
Age
Smoker
non−smoker

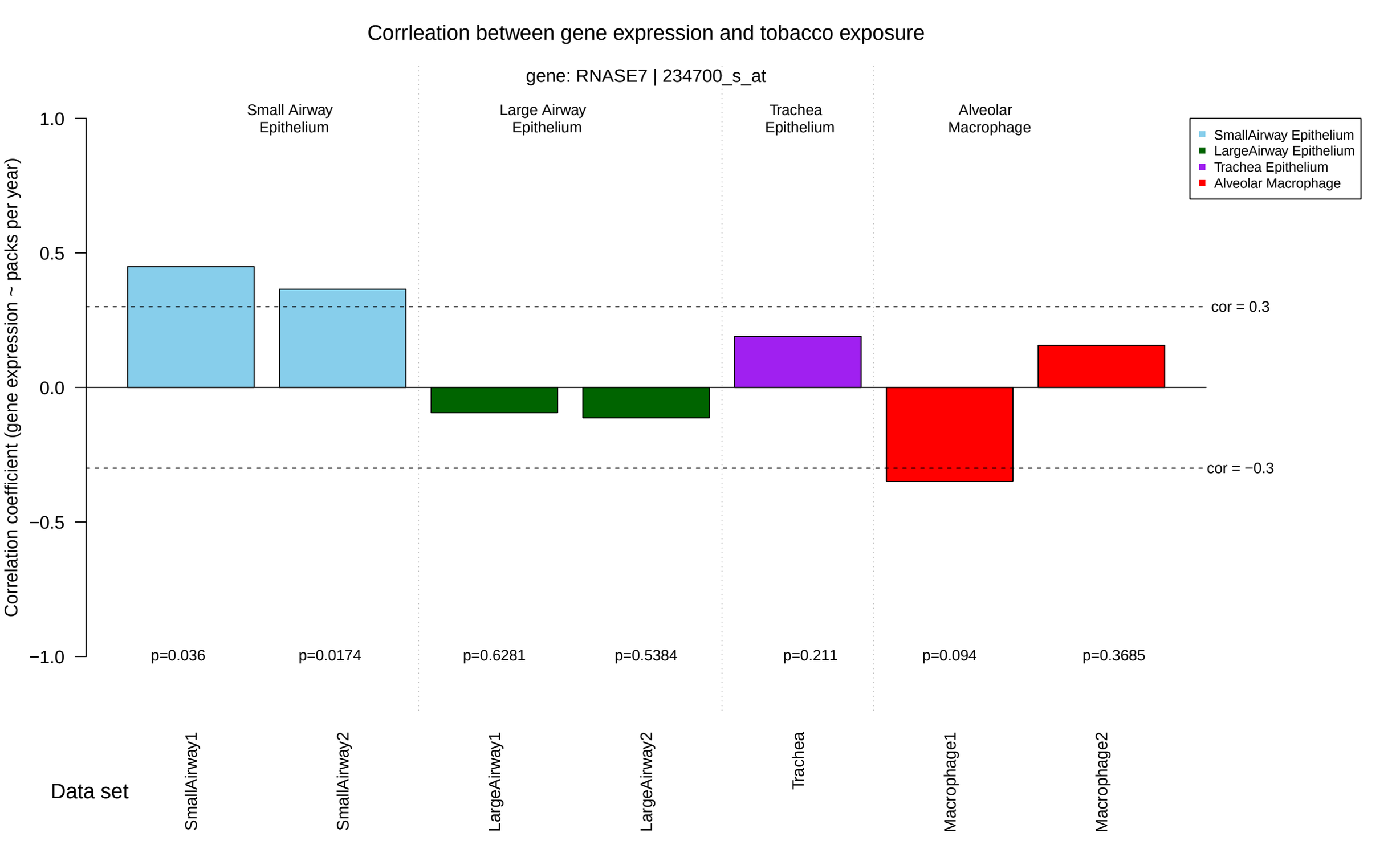

set: LargeAirway1 (p=0.4285)
RNASE7 | 234700_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4748)
RNASE7 | 234700_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6173)
RNASE7 | 234700_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1632)
RNASE7 | 234700_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7023)
RNASE7 | 234700_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNASE7 | 234700_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.036
p=0.0174
p=0.6281
p=0.5384
p=0.211
p=0.094
p=0.3685
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2